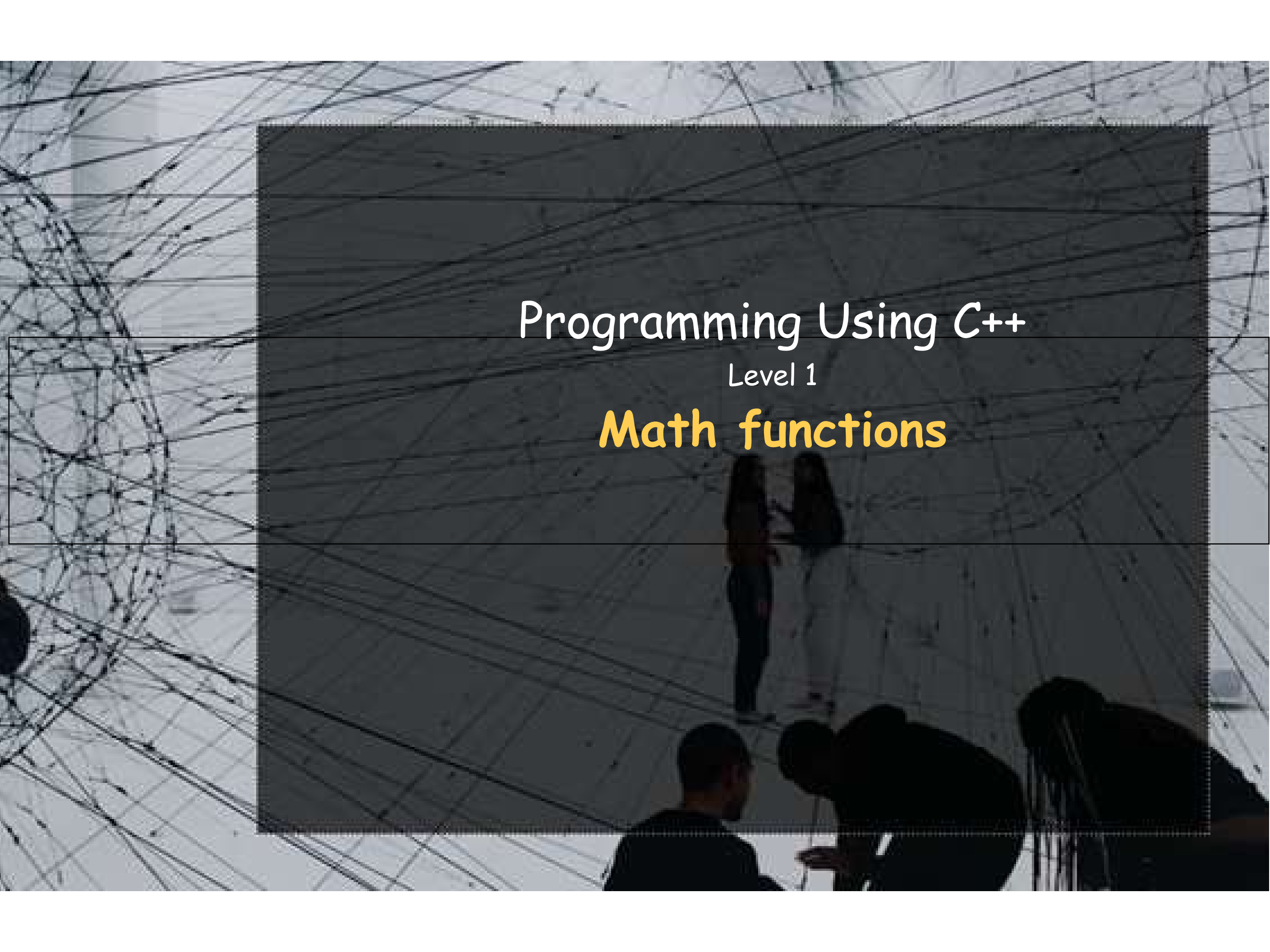


Programming Using C++ Level 1

Math functions



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Math functions

Definition

◆ Math functions are functions provided by the C++ ``<cmath>`` library that are used to perform mathematical calculations.

■ In this program:

■ ``sqrt()`` → calculates the square root.

■ ``pow()`` → raises a number to a power.

Explanation

- ◆ This program calculates the area of a rectangle when one side and the diagonal are known.
- ◆ The diagonal of a rectangle forms a right triangle with its two sides. Therefore, the other side can be calculated using the Pythagorean theorem.
- ◆ The formula used by the program is:
$$\text{Area} = A \times \sqrt{D^2 - A^2}$$
- ◆ ``pow(D, 2)`` calculates D^2 .
- ◆ ``pow(A, 2)`` calculates A^2 .
- ◆ ``sqrt(...)`` calculates the square root of the remaining value.

Steps in main()

1. Read the first side

◆ Store the first side in variable `A`.

2. Read the diagonal

◆ Store the diagonal in variable `D`.

3. Calculate the other side

◆ Calculate:

$\text{sqrt}(\text{pow}(D, 2) - \text{pow}(A, 2))$

4. Calculate the area

◆ Multiply `A` by the calculated second side.

5. Store the result

◆ Store the calculated area in the `float` variable `Area`.

6. Display the result

◆ Print the rectangle area.

💡 Example

◆ Input:

A = 5

D = 40

◆ The program calculates:

Area = $5 \times \sqrt{40^2 - 5^2}$

➡ Result

Rectangle Area = 198.431

Note

- ◆ ``#include <cmath>`` is required because the program uses ``pow()`` and ``sqrt()``.
- ◆ ``float`` is used for ``Area`` because the result can contain decimal values.

⊘ Warning

- ◆ The diagonal `D` must be greater than the side `A` for the expression inside `sqrt()` to produce a valid real-number result.



Comparison

pow()

- ✓ Used to calculate a number raised to a power.
- ✓ ``pow(D, 2)`` means D^2 .
- ✓ Provided by the ``<cmath>`` library.

sqrt()

- ✓ Used to calculate the square root.
- ✓ ``sqrt(25)`` gives 5.
- ✓ Provided by the ``<cmath>`` library.
- ✓ False becomes True.

✦ Summary

- ✓ ``pow()`` calculates powers and ``sqrt()`` calculates square roots.
- ✓ The program uses the Pythagorean theorem to find the missing side of the rectangle.
- ✓ The rectangle area is calculated using the known side and diagonal.

Thank you 😊